Colin Zeng

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Education

Duke University GPA: 3.93/4.0

Bachelor of Science in Computer Science and Statistics *Expected May 2028*

- Relevant Coursework: Machine Learning, Computer Systems, Database Systems, Data Structures and Algorithms, Multivariable Calculus, Linear Algebra, Advanced Probability, Regression Analysis
- Extracurriculars: Catalyst (Duke's Premier Tech Society), Machine Learning, Statistical Science Majors Union
- **Awards:** Ultimate Frisbee U20 World Champion, Team USA, World Junior Ultimate Championships

Experience

Undergraduate Research Intern Jan. 2026 – Present

Dietrich Lab @ Duke University | fungalpangenome.ccn.duke.edu *Durham, NC*

- Designed a PostgreSQL schema mapping 1,482 S. cerevisiae strains across core genes and pangenome traits, serving queries in under 10 ms so researchers can compare gene presence across strains
- Built a full stack web app (React, JavaScript, SQL) for a team of 15 lab researchers, with sitewide search and session based autocomplete caching, released to the department for usage
- Developed the *Ashbya gossypii* Genome Database section in PHP and JavaScript, an SGD style browser with live search across 6,294 genes, presenting project direction to the head of SGD and at the Yeast Genetics Meeting
- Wrote a PHP pipeline parsing ortholog and pathway spreadsheets into per gene JSON records, and integrated BLAST search and a JBrowse2 genome browser, giving Ashbya researchers sequence and genome views

Software Engineer Intern Oct. 2024 - Aug. 2025

Quantworks/SIZEO *Durham, NC*

- Built a web platform on Anthropic's Claude API that generated budget allocation, packaging, and seasonal inventory forecasts for retail clients including JCPenney and Aeropostale, giving merchandise planners recommendations
- Engineered Python data pipelines that ingest client uploaded CSVs and transform them into interactive sales and demand views using Streamlit and Plotly, processing data for client dashboards
- Implemented an artifact store that persists LLM generated charts and reports for retrieval augmented analysis, letting companies revisit and compare past outputs for planned inventory decisions

Projects

Speech Dysfluency Detection and Removal | *Python, Whisper, scikit-learn*

- Built an end-to-end pipeline to detect, categorize, and remove speech dysfluencies (e.g., repetitions)
- Used OpenAI Whisper to transcribe 162 audio clips and rule based logic to flag and categorize 5 dysfluency types (repetition, insertion, deletion, pause, substitution)
- Benchmarked 3 classifiers on an 88 sample test set; a NeMo model reached 79.6% accuracy (0.78 macro F1), a 44% gain over an MFCC random forest baseline and 3 times as accurate as the Llama 3 baseline

Insight, Journal Reflection App | *React, Groq API, LLM Integration*

- Built a React app that generates personalized reflective prompts from journal entries via the Groq API
- Designed retrieval system from scratch using vectorization and cosine similarity to query relevant entries for RAG
- Shipped a new user onboarding flow, 5 selectable themes, and persistent entry management backed by localStorage

Ultimate Frisbee Play Classifier | *PyTorch, YOLOv8, Computer Vision*

- Built a 2 stage computer vision pipeline that classifies 7 ultimate frisbee plays from images by detecting players and the disc, then converted detections into simplified dot-map representations to train a downstream play classifier
- Fine tuned YOLOv8 on a Roboflow dataset, reaching 85% precision and 87% recall across 3 object classes
- Benchmarked ResNet18 for classification, reaching 91.3% accuracy and outperforming custom CNN by 20 points

Event Horizon, Unreal Engine 5 Game | *C++, Unreal Engine 5, Blueprints*

- Built a 3D precision platformer in Unreal Engine 5 (kooling.itch.io), using Blueprints for in-game triggers and UI
- Wrote reusable C++ components with adjustable features, adding sliding and bouncing behavior to platforms

Technical Skills

Languages: Python, Java, SQL, JavaScript, TypeScript, PHP, C++, R, HTML/CSS
Frameworks & Libraries: React, Vite, PyTorch, scikit-learn, YOLOv8, pandas, NumPy, Streamlit, Plotly
Databases & Infrastructure: PostgreSQL, MySQL/MariaDB, Linux server administration, Vercel, Git, Docker
AI & Bioinformatics: Anthropic Claude API, Groq API, Whisper, BLAST+, JBrowse2, AlphaFold